

PHYSICS-INFORMED BUT NOT PHYSICS-CONSISTENT: ERROR GEOMETRY AND SUBSPACE PROJECTION FOR NEURAL AC POWER FLOW

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ABSTRACT

Recent neural power-flow solvers, including emerging foundation models, achieve accurate voltage predictions, yet such accuracy does not necessarily imply physically consistent solutions. Even small voltage errors can yield large AC power-balance residuals, particularly on large-scale grids. We study this accuracy–consistency gap across PIGNN, GridSFM, GridFM-GraphKit, and LUMINA on realistic 2224-bus GBnetwork scenarios, with cross-grid evaluation of GridSFM over 31 systems. Using an SVD basis fitted to training AC power-flow solutions, we find that neural prediction errors contain substantial components outside the dominant solution subspace. To address this mismatch, calibrated solution-subspace projection (CSP) removes off-subspace prediction components, substantially reducing mean power-balance residual from 0.3159 to 0.0550 p.u. for PIGNN-GC and from 0.2687 to 0.0514 p.u. for GridSFM. These results highlight output-error geometry as a critical factor in physics-consistent neural AC power flow.

Index Terms— AC power flow, neural surrogate, physics violation, error geometry, calibration, distribution shift

1. INTRODUCTION

Modern power-system operation increasingly requires evaluating large ensembles of operating conditions. Variable renewable generation, electrification, storage, and spatially concentrated loads increase the variability of power injections and demand [1], while finite transmission capacity makes congestion, curtailment, and corrective redispatch increasingly consequential [2, 3]. In Germany, Redispatch 2.0 uses load-flow-based grid-security analysis to anticipate bottlenecks and coordinate corrective actions across transmission and distribution systems [4]. Related studies also highlight the growing computational demands placed on load-flow analysis under increasingly diverse operating conditions [5]. In nodal electricity markets, transmission constraints shape locational prices and therefore affect both operational and commercial decisions [6]. AC power flow (AC-PF) is consequently a fundamental nonlinear computational primitive, mapping injections and network conditions to bus voltages and network flows needed for system assessment [7]. Repeated AC-PF evaluations arise in time-series and contingency studies [8], while the same nonlinear AC network equations also underlie optimal power flow and

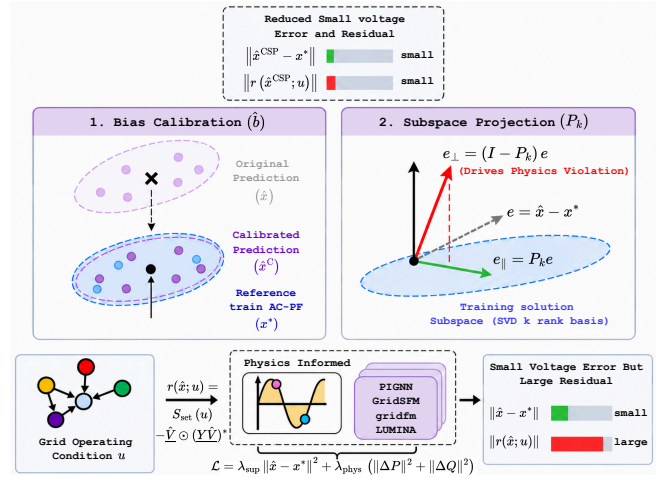


Fig. 1. Neural AC-PF surrogates can achieve small voltage errors while retaining large physics violations. Bias calibration and solution-subspace projection jointly suppress systematic and off-subspace errors, improving both voltage accuracy and AC power-balance consistency.

state estimation [9], where they define operational constraints and nonlinear measurement mappings, respectively. [10].

The computational demands of repeated grid analysis have motivated learned surrogates that amortize the cost of conventional solvers over many operating points [11]. Learned grid solvers span different levels of constraint enforcement: direct regression prioritizes fast inference [12], physics- or constraint-aware objectives penalize violations during training [13, 14], while iterative correction and feasibility-seeking mechanisms refine predictions more directly toward physically consistent or feasible states [15, 16]. More recent cross-grid and foundation-model efforts—including GridSFM [17], GridFM-GraphKit and GENCO [18, 19], and LUMINA [20]—extend this direction with global context and rich heterogeneous, topology-aware representations designed to operate across diverse grids and operating conditions.

Despite this progress, physical consistency remains distinct from state prediction accuracy. Learning-based AC-PF methods commonly regress toward reference solutions obtained from numerical power-flow solvers [11], while physics-informed variants additionally penalize AC power-balance residuals during training [14]. Some formulations go further and optimize the governing residuals directly without

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reference states [15, 21]. Modern evaluations accordingly report both voltage-state errors and reconstructed AC power-balance residuals [14, 19]. However, incorporating residuals into the objective does not enforce the AC equations as hard constraints: non-negligible power-balance violations can remain even when state errors are small and residuals are explicitly minimized. The central question is therefore why such inconsistency persists after both accurate state fitting and physics-aware training. We argue that the missing factor lies in the geometry of the remaining voltage-error pattern across buses.

We investigate this *accuracy–consistency gap* across PIGNN, GridSFM, GridFM-GraphKit, and LUMINA using realistic 2224-bus GBnetwork scenarios and cross-grid evaluation over 31 systems of diverse sizes. Detailed analysis of PIGNN-GC and GridSFM first shows that train-only calibration substantially improves voltage accuracy while leaving non-negligible AC power-balance residuals, indicating that persistent bias alone cannot explain the gap. To characterize the remaining error, we construct an SVD basis exclusively from training AC power-flow solutions and examine how voltage-error patterns align with the resulting training-supported solution subspace. While related PCA-based subspaces have previously been used to compress AC-OPF learning outputs [22], we instead use the subspace as a coordinate system for diagnosing voltage-prediction error relative to training-supported solution variation. We find that substantial error can extend outside this dominant solution subspace, and that the off-subspace component is strongly associated with AC power-balance violation. Motivated by this finding, we introduce *calibrated solution-subspace projection* (CSP), which first removes train-estimated bias and then removes prediction components outside the dominant training-supported solution directions. CSP improves voltage accuracy and AC power balance simultaneously, while controlled distribution shifts show that the useful solution geometry is low-dimensional yet conditioned on the operating distribution. Together, these results identify voltage-error geometry across buses as a key link between state accuracy and physical consistency.

Code: github.com/Kimchangheon/neural-acpf-error-geometry

2. ERROR GEOMETRY AND CALIBRATED SUBSPACE PROJECTION

2.1. AC consistency and train-only calibration

We represent the grid as a directed graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ with $N = |\mathcal{V}|$ buses. Each branch follows the standard π -equivalent model with transformer tap ratios and phase shifts, yielding direction-dependent branch admittances and an asymmetric bus-admittance matrix $\underline{Y} \in \mathbb{C}^{N \times N}$. The voltage state is $x = [V, \theta] \in \mathbb{R}^{2N}$, where $V = |\underline{V}|$ and $\underline{V} = V \odot e^{j\theta}$. Given an operating condition u , comprising nodal setpoints and π -model branch parameters, a neural surrogate predicts $\hat{x} = f_\phi(u)$. Currents are $\underline{I} = \underline{Y}\underline{V}$ and complex powers are $\underline{S} = \underline{V} \odot \underline{I}^*$, giving $P = \Re\{\underline{S}\}$ and $Q = \Im\{\underline{S}\}$. For setpoints $(P^{\text{set}}, Q^{\text{set}})$, define residuals

$$\Delta P = P^{\text{set}} - P, \quad \Delta Q = Q^{\text{set}} - Q.$$

The AC power-flow residual and corresponding nodal power-balance magnitude are

$$r(\hat{x}; u) = \begin{bmatrix} \Delta P \\ \Delta Q \end{bmatrix}, \quad \text{Power Balance (PB)} = \sqrt{\Delta P^2 + \Delta Q^2}.$$

PB is evaluated bus-wise using the standard slack/PV/PQ conventions, and its mean and maximum over buses and scenarios are reported as Mean PB and Max PB. Accordingly, we define AC consistency as

satisfaction of the AC power-balance equations, quantified by PB. All residuals are recomputed from the final predicted voltage state.

Before analyzing error geometry, we apply a train-only calibration that removes post-training bias. For training scenario s , let $\hat{x}_s$ and x_s^* denote the model prediction and reference solution, respectively. The bias is estimated after training and frozen for unseen test scenarios:

$$\hat{b} = \mathbb{E}_{s \in \text{tr}} [\hat{x}_s - x_s^*], \quad \hat{x}^C = \hat{x} - \hat{b}.$$

For phase angles, offsets are estimated using circular differences after consistent gauge alignment. This calibration corrects persistent output bias but does not constrain the directions in which the remaining error varies across buses.

2.2. Error geometry in a training-derived solution subspace

To characterize the remaining error, we construct a low-dimensional basis from Newton–Raphson solutions in the training set. Let $\bar{x}_{\text{tr}}$ denote the training-solution mean and define

$$X = [\bar{x}_1^* - \bar{x}_{\text{tr}} \quad \cdots \quad \bar{x}_{N_{\text{tr}}}^* - \bar{x}_{\text{tr}}] = W \Sigma U^\top, \quad P_k = W_k W_k^\top \quad (1)$$

where P_k denotes the block-diagonal projection onto the dominant rank- k magnitude and gauge-aligned angle subspaces identified from the training data. These subspaces capture dominant solution variation under the training distribution and are therefore distribution dependent. We use them to characterize the geometry of the model's prediction error. For $e = \hat{x} - x^*$, we decompose

$$e^\parallel = P_k e, \quad e^\perp = (I - P_k) e, \quad (2)$$

where $e^\parallel$ lies along the dominant training-solution directions and $e^\perp$ captures error outside them.

To relate this error decomposition to physical consistency, we use a first-order Taylor expansion of the AC residual around a converged power-flow solution x^* :

$$r(x^* + e; u) = \underbrace{r(x^*; u)}_{=0} + \underbrace{J(x^*; u)e}_{\text{first-order term}} + \underbrace{\mathcal{O}(\|e\|^2)}_{\text{higher-order terms}}, \quad (3)$$

where J is the AC residual Jacobian. Thus, the Jacobian determines how a given voltage error is translated into power-balance residual: errors aligned with directions of larger Jacobian gain can produce substantially larger residuals even when their magnitudes are similar. This motivates testing whether AC power balance changes when the voltage error is rotated toward or away from the training-derived solution subspace while its magnitude is held fixed.

2.3. Calibrated solution-subspace projection

This motivates a direct intervention: suppress prediction components outside the dominant training-solution directions while preserving the variation supported by the data. Because systematic bias and directional error are distinct, we first apply the post-training calibration and then project the calibrated prediction:

$$\hat{x}^{\text{CSP}} = \bar{x}_{\text{tr}} + P_k (\hat{x} - \hat{b} - \bar{x}_{\text{tr}}) \quad (4)$$

where $\hat{b}$ is estimated from the trained model's predictions on the training set and frozen at test time, while $\bar{x}_{\text{tr}}$ and P_k are obtained solely from the training reference solutions. Calibration removes systematic output bias, whereas projection removes components outside

the dominant training-solution directions. The two operations are complementary: projection alone can retain the component of the bias lying inside the subspace, whereas CSP removes the bias before restricting the remaining prediction.

CSP is a data-driven output correction rather than a projection onto the nonlinear AC-feasible set, and therefore does not guarantee exact satisfaction of the AC equations or operational constraints. Instead, it provides a direct test of whether removing off-subspace prediction components improves both voltage accuracy and recomputed AC power-balance residuals. All calibration statistics, subspace bases, and rank-selection criteria are determined without using test-set residuals.

3. EXPERIMENTS AND RESULTS

3.1. Experimental setup

Our primary case study is the 2224-bus GBnetwork with 37,022 converged Newton–Raphson operating points, split approximately 1:1:1 into training, validation, and test sets. The data vary system-wide load/generation levels together with bus-level P/Q demand, generator active power, and PV voltage setpoints. We evaluate PIGNN-GC, a global-context extension of PIGNN-Attn-LS [15], GridSFM [17], GridFM-GraphKit [18], and LUMINA [20]; cross-grid breadth is additionally evaluated with GridSFM over 31 systems.

All calibration statistics and solution bases are fitted on training data only. From $k \in \{4, 8, 16, 32, 64\}$, validation Mean PB selects $k^* = 16$ without degrading calibrated voltage RMSE; this rank is then frozen for all test and distribution-shift experiments.

Voltage RMSE alone can be small even when a model largely predicts a static per-bus profile rather than following scenario-dependent voltage variation [17]. To isolate this scenario-dependent response, we center each bus as $\tilde{V}_{is} = V_{is} - \bar{V}_i$ and $\tilde{\hat{V}}_{is} = \hat{V}_{is} - \bar{\hat{V}}_i$, and report

$$R_w^2 = \text{corr}^2(\tilde{V}_{is}, \tilde{\hat{V}}_{is}), \quad a_w = \frac{\text{Cov}(\tilde{V}_{is}, \tilde{\hat{V}}_{is})}{\text{Var}(\tilde{\hat{V}}_{is})}. \quad (5)$$

R_w^2 measures whether the model follows the scenario-to-scenario voltage pattern, while a_w measures how much of the true voltage variation it reproduces. A per-bus constant predictor gives $R_w^2 = a_w = 0$ under our convention, while per-bus calibration leaves both metrics unchanged because the offset vanishes after centering.

3.2. GBnetwork: Error Direction and Calibrated Subspace Projection

Fixed-norm directional intervention. We vary the orientation of the voltage-magnitude error relative to the training-derived solution subspace while holding its ℓ_2 norm fixed, thereby isolating direction from error magnitude. Starting from the calibrated prediction, let

$$\tilde{e}_\perp(\alpha) = \|e_V\|_2 \frac{e_V^\parallel + \alpha e_V^\perp}{\|e_V^\parallel + \alpha e_V^\perp\|_2}, \quad \tilde{e}_\parallel(\alpha) = \|e_V\|_2 \frac{\alpha e_V^\parallel + e_V^\perp}{\|\alpha e_V^\parallel + e_V^\perp\|_2}. \quad (6)$$

By construction, $\|\tilde{e}_\perp(\alpha)\|_2 = \|\tilde{e}_\parallel(\alpha)\|_2 = \|e_V\|_2$. The calibrated phase angle is held fixed, so this experiment changes only the direction of the voltage-magnitude error. We reconstruct

$$\tilde{V}(\alpha) = (V^* + \tilde{e}(\alpha)) \odot e^{j\theta^C} \quad (7)$$

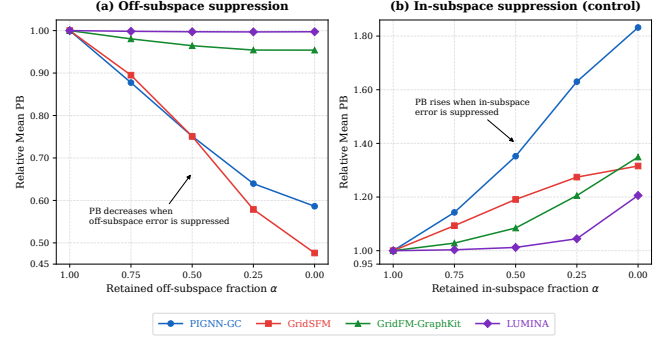


Fig. 2. Fixed-norm directional intervention on GBnetwork. Each curve is normalized by its $\alpha = 1$ Mean PB. Suppressing off-subspace error reduces PB, whereas suppressing in-subspace error produces the complementary increase.

and report

$$\text{Relative PB}(\alpha) = \frac{\text{Mean PB}(\alpha)}{\text{Mean PB}(1)}. \quad (8)$$

Figure 2 directly confirms that residual magnitude depends on error orientation. At $\alpha = 0$, off-subspace suppression lowers Relative Mean PB to 0.586 for PIGNN-GC and 0.476 for GridSFM, whereas suppressing the in-subspace component raises it to 1.832 and 1.316, respectively. GridFM-GraphKit and LUMINA show weaker off-subspace reductions (0.954 and 0.997), but the control intervention again moves PB upward (1.349 and 1.206). Since the voltage-error norm is held fixed, the observed changes isolate the effect of error direction on physical consistency.

Post-training CSP correction. The controlled experiment above requires the reference solution and is used to isolate the role of error direction. We now evaluate CSP directly on the frozen model outputs. Table 3 compares the raw prediction, post-training calibration (C), projection (P_{16}), and their composition, CSP_{16} . CSP projects both voltage magnitude and gauge-aligned phase angle using separate bases derived from the training solutions.

Calibration substantially reduces voltage RMSE but leaves R_w^2 and a_w unchanged, as expected for a per-bus translation. Projection acts on a different component: for PIGNN-GC, CSP_{16} reduces Mean PB from 0.3159 to 0.0550 and Max PB from 195.9 to 3.93, while improving $|V|$ RMSE from 0.0285 to 0.0122. GridSFM shows the same pattern, with Mean PB decreasing from 0.2687 to 0.0514 and $|V|$ RMSE from 0.0287 to 0.0078. GridFM-GraphKit and LUMINA also exhibit large PB reductions, but their much smaller tracking scores show why low PB alone is insufficient to establish a useful scenario-dependent surrogate. The per-bus mean provides the same caution from the opposite direction: it has no scenario tracking ($R_w^2 = a_w = 0$) despite a moderate Mean PB.

3.3. Cross-grid breadth and distribution shift

Cross-grid breadth. Fig. 3 reports GridSFM Mean PB before and after CSP_{16} on 31 systems. The benchmark spans 4–9,241 buses and sub-kV to 750-kV voltage levels, and consists of 28 pandapower/PYPOWER-format systems from IEEE, PEGASE, RTE, and GB benchmark families, plus three Common Information Model (CIM)-based network models; full details are provided in the project repository. CSP_{16} consistently reduces Mean PB across the full range of network sizes, showing that its benefit is not confined to

Table 1. GBnetwork results at $k = 16$ over three seeds (mean $\pm$ standard deviation). C denotes calibration, P_{16} projection, and CSP_{16} their composition. $a_w = 1$ is ideal. The per-bus mean baseline uses training-set means.

Model	Output	$ V $ RMSE $\downarrow$	Angle RMSE ($^\circ$) $\downarrow$	R_w^2 $\uparrow$	a_w	Mean PB $\downarrow$	Max PB $\downarrow$
Per-bus mean	Baseline	.02341	6.847	.000	.000	.12084	5.88
PIGNN-GC	Raw	.02850 $\pm$.00232	1.108 $\pm$.175	.671 $\pm$.050	.546 $\pm$.072	.31592 $\pm$.03240	195.89 $\pm$ 52.54
	C	.01393 $\pm$.00105	.777 $\pm$.161	.671 $\pm$.050	.546 $\pm$.072	.16661 $\pm$.02352	190.00 $\pm$ 37.53
	P_{16}	.01624 $\pm$.00254	1.091 $\pm$.166	.829 $\pm$.045	.541 $\pm$.071	.07594 $\pm$.00422	5.07 $\pm$.85
	CSP_{16}	.01222 $\pm$.00134	.835 $\pm$.141	.829 $\pm$.045	.541 $\pm$.071	.05497 $\pm$.00257	3.93 $\pm$.07
GridSFM	Raw	.02868 $\pm$.01652	2.294 $\pm$.302	.768 $\pm$.151	.724 $\pm$.149	.26867 $\pm$.13004	9.10 $\pm$ 4.02
	C	.01086 $\pm$.00403	1.781 $\pm$.055	.768 $\pm$.151	.724 $\pm$.149	.08251 $\pm$.01093	7.29 $\pm$ 3.53
	P_{16}	.01814 $\pm$.01145	2.244 $\pm$.325	.936 $\pm$.032	.724 $\pm$.149	.07630 $\pm$.01942	5.08 $\pm$.49
	CSP_{16}	.00782 $\pm$.00307	1.807 $\pm$.053	.936 $\pm$.032	.724 $\pm$.149	.05136 $\pm$.00143	4.94 $\pm$.42
GridFM-GraphKit	Raw	.03695 $\pm$.00162	5.431 $\pm$.632	.208 $\pm$.085	.151 $\pm$.040	2.20492 $\pm$.41713	4651.17 $\pm$ 1620.49
	C	.02107 $\pm$.00090	3.108 $\pm$.503	.208 $\pm$.085	.151 $\pm$.040	1.39869 $\pm$.36963	4853.45 $\pm$ 1595.78
	P_{16}	.02702 $\pm$.00102	3.111 $\pm$.582	.591 $\pm$.128	.147 $\pm$.040	.12340 $\pm$.01484	6.68 $\pm$.32
	CSP_{16}	.02016 $\pm$.00092	2.055 $\pm$.431	.591 $\pm$.128	.147 $\pm$.040	.07598 $\pm$.00290	4.83 $\pm$.36
LUMINA	Raw	.04355 $\pm$.00094	17.068 $\pm$.106	.059 $\pm$.043	.047 $\pm$.035	138.813 $\pm$ 12.092	24178.96 $\pm$ 6886.75
	C	.02273 $\pm$.00050	6.424 $\pm$.097	.059 $\pm$.043	.047 $\pm$.035	2.09892 $\pm$.52156	4063.46 $\pm$ 2201.01
	P_{16}	.02959 $\pm$.00078	13.414 $\pm$.318	.445 $\pm$.034	.045 $\pm$.033	.25953 $\pm$.02210	12.88 $\pm$.22
	CSP_{16}	.02240 $\pm$.00075	6.147 $\pm$.172	.445 $\pm$.034	.045 $\pm$.033	.10101 $\pm$.00359	5.45 $\pm$.15

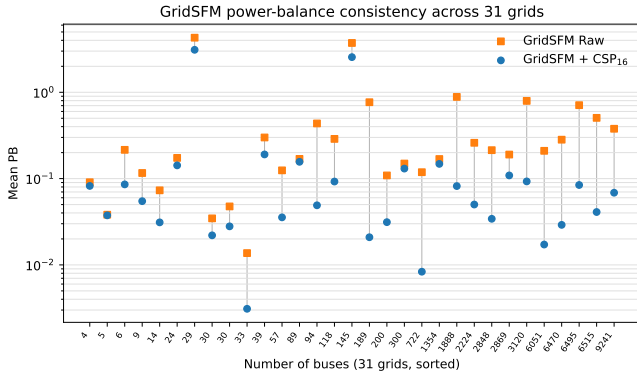


Fig. 3. Grid-by-grid physical consistency of GridSFM before and after CSP_{16} across 31 systems ordered by bus count. Mean PB is shown on a logarithmic scale.

GBnetwork.

Topology and spatial shifts. We next keep a PIGNN-GC checkpoint and $k = 16$ fixed while changing the evaluation distribution. For $N-1$ and $N-2$, the control basis is transferred without modification to unseen line-outage topologies. GBcorr instead redistributes load across connected zones using a GB-derived spatial covariance while preserving total system load. We evaluate both direct transfer of the control calibration and solution basis and condition-matched refitting using GBcorr training data only; model weights and rank remain unchanged.

Table 2 reveals two qualitatively different failure modes. Under $N-1$ and $N-2$ outages, voltage-magnitude tracking remains close to control ($R_w^2 \approx 0.84$), and the bulk of the PB distribution changes little: Mean PB rises from 0.0581 to 0.0603 and 0.0617, while p99 moves only from 0.5462 to 0.5632 and 0.5721. The maxima, however, increase sharply to 33.16 and 27.45, while the bulk statistics remain close to control, indicating that topology shift primarily introduces isolated extreme violations.

Table 2. Fixed-rank CSP_{16} under topology and spatial distribution shifts. It uses separate condition sets from model training; PB means and quantiles are pooled over all bus-scenario pairs.

Condition	RMSE $_V$	R_w^2 $\uparrow$	a_w	Mean	Median	p95	p99	Max
Control	.01281	.8476	.4866	.0581	.0131	.2665	.5462	3.58
$N-1$ (T)	.01314	.8391	.4750	.0603	.0131	.2713	.5632	33.16
$N-2$ (T)	.01288	.8458	.4870	.0617	.0132	.2713	.5721	27.45
GBcorr (T)	.01966	.4819	.4025	.1003	.0231	.4512	.9746	7.00
GBcorr (R)	.01620	.5150	.4247	.0680	.0174	.3031	.6415	4.99

T: transfer of the control basis; R: basis refitted using shifted training solutions only.

GBcorr produces a broad rather than tail-dominated shift. R_w^2 drops from 0.8476 to 0.4819, while Mean PB rises to 0.1003 and p99 to 0.9746; the median and p95 increase as well. Condition-matched recalibration and subspace refitting reduces Mean PB to 0.0680 and p99 to 0.6415, while improving $|V|$ RMSE. Tracking recovers only partially ($R_w^2 = 0.5150$), indicating that basis adaptation can improve physical consistency, whereas recovering scenario-dependent tracking under this shift may require updating the surrogate parameters using data from the shifted operating distribution.

4. CONCLUSION

This work advances the understanding of neural AC power-flow surrogates by identifying output-error geometry as a key source of the gap between voltage accuracy and physical consistency. Across physics-informed and foundation-model approaches, we show that small voltage errors can contain off-subspace components that induce large AC power-balance violations. Building on this insight, calibrated solution-subspace projection removes off-subspace prediction components, reducing both voltage error and AC power-balance residuals across diverse grids. These results motivate extending physics-consistent neural surrogates toward broader power-system analysis, estimation, and optimization.

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